

Ehsan Barkhordar

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Education

Koç University (Ranked #1 in Turkey, QS 2026)

M.Sc. in Computer Science, GPA: 3.8/4.0

Feb 2023 – Feb 2026

Istanbul, Turkey

- Joined the Ph.D. program in Feb 2023; transferred to the M.Sc. (with thesis) in Apr 2025.
- Advisors: Prof. Gözde Gül Şahin (primary), Assoc. Prof. Erdem Yörük (co-advisor).

Amirkabir University of Technology (Ranked #3 in Iran, QS 2026)

M.Sc. in Computer Science, GPA: 15.15/20 (3.1/4.0)

Sep 2018 – Mar 2021

Tehran, Iran

Amirkabir University of Technology (Ranked #3 in Iran, QS 2026)

B.Sc. in Computer Science, GPA: 14.79/20 (3.0/4.0)

Sep 2013 – Apr 2018

Tehran, Iran

Publications

Are Non-English Papers Reviewed Fairly? Language-of-Study Bias in NLP Peer Reviews

- Ehsan Barkhordar, Abdulfattah Sifa, Verena Blaschke, Erika Lombart, Marie-Catherine de Marneffe, Gözde Gül Şahin. *Proceedings of EMNLP 2026 (Main Conference)*, 2026, Budapest, Hungary. To appear.

Assessing the Predictive Power of Social Media Data-Fed Large Language Models on Vote Preference

- Ehsan Barkhordar, Şükrü Atsızelti. *Companion Publication of the 16th ACM Web Science Conference*, 2024, Stuttgart, Germany.

Why the Unexpected? Dissecting the Political and Economic Bias in Persian Small and Large Language Models

- Ehsan Barkhordar, Surendrabikram Thapa, Ashwarya Maratha, Usman Naseem. *Proceedings of the 3rd Annual Meeting of SIGUL @ LREC-COLING*, 2024, Torino, Italy.

Team Curie at HSD-2Lang 2024: Hate Speech Detection in Turkish and Arabic Tweets using BERT-based Models

- Ehsan Barkhordar, Işık Topçu, Ali Hürriyetoğlu. *Proceedings of the 7th Workshop on CASE*, 2024, St. Julian's, Malta.

Which Side Are You On? Investigating Politico-Economic Bias in Nepali Language Models

- Surendrabikram Thapa, Kritesh Rauniyar, Ehsan Barkhordar, Hariram Veeramani, Usman Naseem. *Proceedings of ALTA*, 2024, Canberra, Australia.

Clustering of Bank Customers using LSTM-based Encoder-Decoder and Dynamic Time Warping

- Ehsan Barkhordar, Mohammad Hassan Shiralil-Shahreza, Hamid Reza Sadeghi. *arXiv preprint arXiv:2110.11769*, 2021.

Research & Work Experience

Koç University

Istanbul, Turkey

Research Project Engineer, with Prof. Zeynep Akşin Karaesmen

Mar 2026 – Sep 2026

- Built a text-mining pipeline over a large X (Twitter) corpus to extract the Hurricane Helene / FEMA misinformation narrative and split it by diffusion phase, as input to network analysis and an agent-based simulation of how misinformation spreads.

Researcher in NLP, with Prof. Gözde Gül Şahin

Feb 2023 – Feb 2026

- First-author EMNLP 2026 main-conference paper on language-of-study bias in NLP peer review, with an LLM-based detection pipeline over 15,645 reviews.
- Represented procedural text as executable Python code to guide LLM reasoning, raising accuracy from 68.29% to 86.76%.
- Supervised two research interns and reviewed 100+ papers for ARR, EMNLP, and ICLR.

Researcher in Computational Social Science, with Assoc. Prof. Erdem Yörük

Mar 2024 – Sep 2024

- Built *Yildirim*, an annotation platform for Turkish tweets, annotated 5,000 tweets from a 700M+ tweet dataset, and fine-tuned BERT and RoBERTa for named entity recognition.
- Tested whether LLMs fed with social media data can predict vote preference (first-author, ACM WebSci 2024).
- Built BERT-based hate speech classifiers for Turkish and Arabic tweets (first-author, CASE 2024).

Enlighty

Technical Lead

Sep 2023 – Feb 2026

Remote

- Led the technical side of an AI consumer-analytics platform (emotion detection, brand health) with a FastAPI/ClickHouse backend, Nuxt frontends, and an LLM assistant, in a team of 12 engineers.
- Built *Enlighter*, an LLM agent that answers analytics questions by calling 83 API endpoints (LangChain, OpenAI tool calling), with streamed responses and PDF reports.
- Reduced API latency with ClickHouse connection pooling and non-blocking async queries.
- Containerized all four services, built GitHub Actions CI/CD, and added multi-region ClickHouse routing.

Pishro Net Energy

Machine Learning Engineer

Aug 2022 – Jan 2023

Tehran, Iran

- Built energy forecasting models on 1,000+ real-time sensor streams across 10 buildings, reducing forecasting error by 21%.
- Built dashboards tracking 150+ KPIs for facility managers.

Sokan Platform

Machine Learning Engineer

Tehran, Iran

Apr 2021 – Feb 2022

- Built churn prediction (82% accuracy) and cohort-based sales models.
- Built data pipelines over 1B+ records with TensorFlow and Kubeflow, including Mini-batch K-means clustering.
- Used FP-Growth association mining to suggest product bundles for cross-selling.

Senior Software Engineer

Sep 2020 – Apr 2021

- Moved long-running Django tasks to Celery workers, reducing compute overhead by 40%.
- Migrated analytics queries to ClickHouse, making them 5x faster.
- Built 50+ GraphQL APIs with Graphene, replacing several REST endpoints.
- Built Docker/GitLab CI/CD pipelines, cutting deployment time by 60% and tripling release frequency.

Bale Messenger

Software Engineer

Jan 2018 – Feb 2020

Tehran, Iran

- Worked on a cross-platform messaging app with 8M+ users.
- Built a Python bot SDK used by 200+ developers to run 500+ bots.
- Helped migrate the backend to Go microservices, making it 7x faster.
- Set up Prometheus and Grafana monitoring for 20+ system metrics.

Selected Projects

LLM Collusion and Identifiability in Code Generation | May 2025 – Present

- Investigates whether large language models can recognize or attribute code authored by themselves or other LLMs.
- Implemented three attribution tasks: Self-Recognition, Target Identification, and Full Attribution.
- Evaluated across multiple benchmarks (MBPP, HumanEval, DS-1000) with code generated by GPT-5, Claude, Gemini, Grok, and DeepSeek.
- Fine-tuned lightweight judge models to identify model-specific code signatures, highlighting risks of collusion and leakage in code evaluation.

Multimodal Framework for Detecting Implicit Hate Speech on Social Media | Sept 2024 – Apr 2025

- Designed a hybrid framework combining text encoders (BERT, LLaMA) with vision backbones (Swin Transformer, ViT) to detect implicit hate speech in image-text posts.
- Improved robustness via focal loss, progressive unfreezing, and adversarial training.

Intent Prediction and Slot Filling with Alexa MASSIVE Dataset | Feb 2022 – Apr 2022

- Participated in the Massively Multilingual NLU 2022 shared task.
- Trained multilingual intent and slot-filling models on the Alexa MASSIVE dataset using PyTorch.

Shilling Attack Detection in Recommender Systems | May 2020

- Investigated detection of profile injection (shilling) attacks in recommender systems.
- Extracted RDMA, WDA, and TF-IDF features; evaluated multiple classifiers (KNN, Random Forest, SVM).

Teaching Experience

Koç University

Fall 2024 – Spring 2025

Teaching Assistant – COMP 302: Software Engineering (2 consecutive terms)

Istanbul, Turkey

- Ran labs on software testing, CI/CD, and agile methods.
- Wrote grading rubrics, graded semester projects, and helped 60+ students with Git and team workflows.

Koç University

Spring 2024

Teaching Assistant – COMP 100: Introduction to CS and Programming

Istanbul, Turkey

- Assisted an introductory Python course with 200+ students: graded weekly assignments, ran labs, and helped students debug their code.

Koç University

Fall 2023

Teaching Assistant – COMP 491: Senior Design Project

Istanbul, Turkey

- Advised senior students on project design and implementation, and graded their reports and presentations.

Koç University

Spring 2023

Teaching Assistant – COMP 202: Data Structures and Algorithms in Java

Istanbul, Turkey

- Ran weekly labs on data structures in Java for 50+ students, held office hours, and graded assignments and projects.

Honors and Awards

2nd Rank – Hate Speech Detection in Turkish and Arabic Tweets (HSD-2Lang) 2024

Invited Participant – Lisbon Machine Learning School (LxMLS 2023), Lisbon, Portugal 2023

One of 5 Selected Teams – Fusion Startup Event (sponsored by divar.ir) 2022

28th Place / 1608 Participants – Django/Python Code Cup, quera.ir 2020

2nd Team – Hackathon, Amirkabir University 2019

Winner – BOT-CUP, Sharif University of Technology 2018

Skills

Programming: Python, SQL, Bash

ML & NLP: PyTorch, Hugging Face (Transformers, Datasets, Accelerate, PEFT), scikit-learn, Pandas, NumPy

LLMs: Fine-tuning, LLM evaluation, tool-calling agents (LangChain, OpenAI), text classification, NER, multilingual NLP

Research Tools: LaTeX, Git/GitHub, Jupyter, MLflow, Weights & Biases

Engineering: FastAPI, Django, Celery, GraphQL, ClickHouse, Docker, GitHub Actions

Languages

Persian: Native

English: Fluent – TOEFL iBT 96/120 (test centre, Dec 2024) and 103/120 (Home Edition, Dec 2024)

Turkish: Beginner